

Coding & Solution

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

This section documents the implemented Rising Waters codebase, demonstrating working functionality and alignment with the defined requirements.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Pallaanilkumar655/Rising-Waters-Flood-Prediction
Programming Language(s)	Python
Framework(s) Used	Flask
Key Features Implemented	- Random Forest based flood prediction - Flask web app with form-based input - Real-time prediction display - Retractable model pipeline (train_model.py)
Pending / Incomplete Features	Live weather API integration, cloud deployment, mobile app version
Setup / Run Instructions	1. git clone the repository 2. pip install -r requirements.txt 3. python train_model.py 4. python app.py 5. Open http://127.0.0.1:5000

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Partial – basic handling in place
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

The model was trained using RandomForestClassifier from scikit-learn and serialized with Joblib for fast loading inside the Flask application at runtime.